API and Database Reference

AI Career & Skill Roadmap Platform. Spring Boot gateway APIs, Node.js worker APIs, and every database table with its columns.

How the two backends split the work

Backend	Owns	Talks to	Notes
Spring Boot gateway	Authentication, users, organizations, all CRUD, skill graph, item bank, adaptive test engine, task tracking, the only database	Web app, mobile app (public). Node worker (private).	The only service clients ever call. Validates every request, stores everything in PostgreSQL.
Node.js AI worker	Slow, asynchronous LLM jobs: roadmap generation, question drafting, result explanations. Produces validated JSON.	Gateway only. LLM provider. Redis queue.	Stateless, no database access. Reachable only on the private network with a shared secret header.

Conventions

Topic	Rule
Public base path	/api/v1 (gateway). All responses are JSON.
Internal base path	/internal on both services. Requires header X-Internal-Key. Never exposed to the internet.
Authentication	Authorization: Bearer <access token>. Access token lives about 15 minutes, refresh token 7 to 30 days and is rotated on each use.
Roles	System roles: USER, PLATFORM_ADMIN (authoring endpoints). Organization roles: OWNER, ADMIN, MENTOR, LEARNER.
Errors	{ "error": { "code": "...", "message": "...", "details": [] } } with standard HTTP status codes (400, 401, 403, 404, 409, 422, 429, 500).
Pagination	?page=0&size=20 on list endpoints. Response: { items, page, size, totalItems }.
IDs and times	UUIDs for all ids. Timestamps are ISO 8601 UTC.
Phase column	S1 = build in Sprint 1. S1 opt = nice to have in Sprint 1. Later = planned for Sprint 2 and beyond (needs the Node worker or organizations).

Part 1. Spring Boot gateway APIs

Public API used by the web and mobile apps. Auth column: Public = no token, JWT = any logged-in user, Owner = JWT and the record belongs to the caller, Admin = PLATFORM_ADMIN, Org role = member of that organization with the stated role.

1.1 Authentication

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/auth/signup	Public	S1	Create an account	{ email, password, fullName }	201 { user, accessToken, refreshToken, expiresIn }
POST	/auth/login	Public	S1	Log in with email and password	{ email, password }	200 { user, accessToken, refreshToken, expiresIn }
POST	/auth/refresh	Refresh token	S1	Exchange refresh token for a new pair (rotation)	{ refreshToken }	200 { accessToken, refreshToken, expiresIn }
POST	/auth/logout	JWT	S1	Revoke the refresh token	{ refreshToken }	204
POST	/auth/forgot-password	Public	S1 opt	Send a password reset link	{ email }	202 (always, to avoid leaking accounts)
POST	/auth/reset-password	Public	S1 opt	Set a new password using the emailed token	{ token, newPassword }	204

Login failures increment failed_login_count and lock the account for a few minutes after repeated failures (429 while locked).

1.2 Current user

Method	Endpoint	Auth	Phase	Purpose	Request	Response
GET	/me	JWT	S1	Profile of the logged-in user	-	200 { id, email, fullName, systemRole, organizations[] }
PATCH	/me	JWT	S1	Update display name	{ fullName }	200 updated user
POST	/me/change-password	JWT	S1 opt	Change password while logged in	{ currentPassword, newPassword }	204
DELETE	/me	JWT	Later	Delete own account and data	-	204

1.3 Tracks and skill graph (read)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
GET	/tracks	JWT	S1	List tracks; non-available ones return COMING_SOON	-	200 [{ id, slug, name, description, icon, status }]
GET	/tracks/{trackId}	JWT	S1	One track	-	200 track
GET	/tracks/{trackId}/graph	JWT	S1	All skills and prerequisite edges for the skill map	-	200 { skills: [{ id, name, cluster, difficulty }], edges: [{ skillId, prerequisiteId }] }

1.4 Diagnostic assessment (adaptive test)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/assessments	JWT	S1	Start a test for a track. If one is already in progress, returns it instead.	{ trackId }	201 { assessmentId, question, progress: { answered, max } }
GET	/assessments/{id}	Owner	S1	Status of a test; current question if still in progress (used to resume)	-	200 { id, status, question?, progress }
POST	/assessments/{id}/responses	Owner	S1	Submit an answer. Server scores, updates ability, picks the next question or finishes.	{ itemId, selectedOption, responseTimeMs? }	200 { done: false, question, progress } or { done: true }
GET	/assessments/{id}/result	Owner	S1	Final result of a completed test	-	200 { level, theta, se, ciLow, ciHigh, answered, skills: [{ skillId, state }] }
GET	/assessments	JWT	S1	Own test history (dashboard)	?trackId, ?status, ?page, ?size	200 page of { id, track, level, status, startedAt, completedAt }
POST	/assessments/{id}/abandon	Owner	S1	Abandon an in-progress test	-	204

Question objects contain only { itemId, stem, options[] }. The correct answer and IRT values are never sent to the client.

1.5 Dashboard

Method	Endpoint	Auth	Phase	Purpose	Request	Response
GET	/dashboard	JWT	S1	One call for the dashboard: level per track, recent tests, pending AI tasks	-	200 { tracks: [{ track, level, ci, lastAssessmentAt }], recentAssessments[], pendingTasks }

1.6 Admin authoring (skill graph and question bank CRUD)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/admin/tracks	Admin	S1	Create a track	{ slug, name, description, status }	201 track
PATCH	/admin/tracks/{trackId}	Admin	S1	Edit a track or change its status	{ name?, description?, status?, displayOrder? }	200 track
POST	/admin/tracks/{trackId}/skills	Admin	S1	Add a skill	{ slug, name, cluster, difficulty, description }	201 skill
PATCH	/admin/skills/{skillId}	Admin	S1	Edit a skill	{ name?, cluster?, difficulty?, description? }	200 skill
DELETE	/admin/skills/{skillId}	Admin	S1	Delete a skill (blocked if used by items)	-	204
PUT	/admin/skills/{skillId}/prerequisites	Admin	S1	Replace the prerequisite list. Rejects cycles.	{ prerequisiteIds: [] }	200 skill; 422 if a cycle would form
POST	/admin/items	Admin	S1	Create a question	{ trackId, stem, options[], correctOption, explanation?, cluster, irtA, irtB, irtC, skillIds[] }	201 item
GET	/admin/items	Admin	S1	List questions	?trackId, ?status, ?cluster, ?page	200 page of items (includes answers and IRT values)
PATCH	/admin/items/{itemId}	Admin	S1	Edit a question or move DRAFT to ACTIVE	any item field	200 item
DELETE	/admin/items/{itemId}	Admin	S1	Retire a question (soft: status RETIRED)	-	204
PUT	/admin/items/{itemId}/skills	Admin	S1	Replace the skills a question is tagged with	{ skillIds: [] }	200 item
POST	/admin/tracks/{trackId}/import	Admin	S1	Bulk import skills, edges and items from YAML or JSON; validates DAG	multipart file	200 { skillsCreated, itemsCreated, errors[] }

1.7 AI tasks (gateway tracks them and forwards to the Node worker)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/tasks	JWT	Later	Create an async AI task. Gateway validates input, saves it as PENDING, forwards to the worker.	{ type, input, idempotencyKey? }	202 { taskId, status: PENDING }
GET	/tasks/{taskId}	Owner	Later	Poll a task	-	200 { id, type, status, progress, result?, error? }
GET	/tasks	JWT	Later	List own tasks	?type, ?status, ?page	200 page of tasks
POST	/tasks/{taskId}/cancel	Owner	Later	Cancel a PENDING or RUNNING task	-	202

1.8 Roadmaps (created from AI tasks)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/roadmaps/generate	JWT	Later	Start roadmap generation from a completed assessment (creates a ROADMAP_GENERATE task)	{ assessmentId, goals? }	202 { taskId }
GET	/roadmaps	JWT	Later	List own roadmaps	?trackId, ?status	200 page of roadmaps
GET	/roadmaps/{roadmapId}	Owner	Later	Roadmap with all steps	-	200 { roadmap, steps[] }
PATCH	/roadmaps/{roadmapId}/steps/{stepId}	Owner	Later	Mark a step in progress or done	{ status }	200 step
DELETE	/roadmaps/{roadmapId}	Owner	Later	Archive a roadmap	-	204

1.9 Organizations and members

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/organizations	JWT	Later	Create an organization (caller becomes OWNER)	{ name, slug, description? }	201 organization
GET	/organizations	JWT	Later	Organizations the caller belongs to	-	200 [organization]
GET	/organizations/{orgId}	Org member	Later	One organization	-	200 organization
PATCH	/organizations/{orgId}	Org ADMIN+	Later	Edit name or description	{ name?, description? }	200 organization
DELETE	/organizations/{orgId}	Org OWNER	Later	Archive the organization	-	204
GET	/organizations/{orgId}/members	Org member	Later	List members and roles	?page	200 page of members
PATCH	/organizations/{orgId}/members/{userId}	Org ADMIN+	Later	Change a member's role	{ role }	200 member
DELETE	/organizations/{orgId}/members/{userId}	Org ADMIN+	Later	Remove a member	-	204
POST	/organizations/{orgId}/invites	Org ADMIN+	Later	Invite by email	{ email, role }	201 invite
POST	/invites/accept	JWT	Later	Accept an invite	{ token }	200 organization
GET	/organizations/{orgId}/assessments	Org MENTOR+	Later	Mentor view of members' test results	?userId, ?trackId, ?page	200 page of assessments

1.10 Gateway internal endpoints (called only by the Node worker)

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/internal/tasks/{taskId}/progress	Internal key	Later	Worker reports progress	{ progress }	204
POST	/internal/tasks/{taskId}/result	Internal key	Later	Worker reports final outcome. Gateway re-validates JSON, stores it, marks DONE or FAILED.	{ status, result?, schemaVersion, errorCode?, errorMessage? }	204

1.11 System

Method	Endpoint	Auth	Phase	Purpose	Request	Response
GET	/health	Public	S1	Liveness check for deployment	-	200 { status: UP }

Part 2. Node.js AI worker APIs

Private service. Only the gateway can call it (private network plus X-Internal-Key). It has no database: it queues jobs, calls the LLM, validates the JSON against the shared schemas in /contracts, and reports back to the gateway. Sprint 1 does not need it, because the diagnostic test uses no LLM.

2.1 Worker endpoints

Method	Endpoint	Auth	Phase	Purpose	Request	Response
POST	/internal/tasks	Internal key	Later	Accept a job and put it on the queue	{ taskId, type, userId, input, maxAttempts? }	202 { jobId, status: QUEUED }
GET	/internal/tasks/{taskId}	Internal key	Later	Job status (fallback if a callback was lost)	-	200 { status, progress, attempts }
POST	/internal/tasks/{taskId}/cancel	Internal key	Later	Cancel a queued or running job	-	202
GET	/internal/task-types	Internal key	Later	Supported task types and schema versions	-	200 [{ type, schemaVersion }]
GET	/internal/health	Internal key	Later	Health, queue depth, LLM reachability	-	200 { status, queueDepth, llmReachable }

2.2 Task types the worker runs

Task type	Input (from gateway)	Output JSON (validated)	Used by	Typical duration
ROADMAP_GENERATE	{ trackId, level, theta, skillStates[], goals?, hoursPerWeek? }	roadmap.schema.json: { title, summary, steps: [{ skillId, title, description, resources[], estimatedHours }] }	POST /roadmaps/generate	15 to 60 s
RESULT_EXPLAIN	{ assessmentId, level, skillStates[] }	{ summary, strengths[], gaps[] }	Result screen (optional)	5 to 20 s
ITEM_DRAFT	{ trackId, skillId, count, difficultyHint }	assessment-item.schema.json drafts for admin review (never auto-activated)	Admin authoring	20 to 90 s
SKILL_GRAPH_DRAFT	{ trackName, targetSkillCount }	Draft DAG { skills[], edges[] } for admin review	Admin authoring	30 to 120 s

2.3 Worker rules

Rule	Detail
Idempotency	The gateway passes taskId. Submitting the same taskId twice does not create a second job.
Retries	Failed LLM calls or invalid JSON are retried up to maxAttempts (default 3) with backoff. After that the task is reported FAILED with an error code.
Validation	Every result is checked against its JSON schema from /contracts before it is sent. The gateway checks it again before saving.
Callbacks	Worker calls gateway /internal/tasks/{taskId}/progress and /result. Clients never talk to the worker and never receive its address.
Error codes	LLM_TIMEOUT, LLM_RATE_LIMIT, INVALID_OUTPUT, INPUT_INVALID, CANCELLED, INTERNAL.

Part 3. Database tables and columns (PostgreSQL)

Owned by the Spring Boot gateway and created with Flyway migrations. Defaults to gen_random_uuid() for ids and now() for created_at. The ai_tasks table and the Later tables are only used once the Node worker and roadmaps arrive. Extension needed: citext.

3.1 Users and authentication

users [S1]

One row per account.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
email	citext	NO	UNIQUE	Login email, case-insensitive
password_hash	text	NO	-	BCrypt hash. Never returned by any API.
full_name	text	NO	-	Display name
system_role	text	NO	default 'USER'; CHECK IN ('USER','PLATFORM_ADMIN')	Platform-wide role. PLATFORM_ADMIN may use /admin endpoints.
status	text	NO	default 'ACTIVE'; CHECK IN ('ACTIVE','DISABLED')	Disabled accounts cannot log in
failed_login_count	smallint	NO	default 0	Consecutive failed logins; reset to 0 on success
locked_until	timestamptz	YES	-	Temporary lock after too many failed logins
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)
last_login_at	timestamptz	YES	-	Time of the latest successful login

Indexes and constraints: UNIQUE (email).

refresh_tokens [S1]

One row per issued refresh token. Rotated on every refresh.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
user_id	uuid	NO	FK users(id) ON DELETE CASCADE	Token owner
token_hash	text	NO	UNIQUE	SHA-256 of the token. The raw token is never stored.
device	text	YES	-	Label such as 'Chrome on Linux' or 'Android app'
user_agent	text	YES	-	Raw user agent string
ip_address	inet	YES	-	IP at creation time
created_at	timestamptz	NO	default now()	Row creation time
expires_at	timestamptz	NO	-	When the token stops working
revoked_at	timestamptz	YES	-	Set on logout or when rotated
replaced_by	uuid	YES	FK refresh_tokens(id)	Next token in the rotation chain (detects token reuse)

Indexes and constraints: INDEX (user_id); UNIQUE (token_hash).

password_reset_tokens [S1 opt]

Single-use tokens for the forgot-password flow.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
user_id	uuid	NO	FK users(id) ON DELETE CASCADE	Account being reset
token_hash	text	NO	UNIQUE	SHA-256 of the emailed token
created_at	timestamptz	NO	default now()	Row creation time
expires_at	timestamptz	NO	-	Usually 30 to 60 minutes after creation
used_at	timestamptz	YES	-	Set when the token is used; a used token is rejected

Indexes and constraints: INDEX (user_id).

3.2 Organizations

organizations [Later (table now)]

Colleges, companies or teams that group learners and mentors.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
name	text	NO	-	Display name
slug	text	NO	UNIQUE	URL-safe short name
description	text	YES	-	Optional about text
created_by	uuid	YES	FK users(id) ON DELETE SET NULL	User who created it
status	text	NO	default 'ACTIVE'; CHECK IN ('ACTIVE','ARCHIVED')	Archived organizations are read-only
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)

organization_members [Later (table now)]

Which users belong to which organization and in what role.

Column	Type	Null	Default / constraint	Description
organization_id	uuid	NO	PK part; FK organizations(id) ON DELETE CASCADE	Organization
user_id	uuid	NO	PK part; FK users(id) ON DELETE CASCADE	Member
role	text	NO	CHECK IN ('OWNER','ADMIN','MENTOR','LEARNER')	Role inside this organization
invited_by	uuid	YES	FK users(id) ON DELETE SET NULL	Who invited this member
joined_at	timestamptz	NO	default now()	When the user joined

Indexes and constraints: PRIMARY KEY (organization_id, user_id); INDEX (user_id).

organization_invites [Later (table now)]

Pending email invitations.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
organization_id	uuid	NO	FK organizations(id) ON DELETE CASCADE	Target organization
email	citext	NO	-	Invited email address
role	text	NO	CHECK IN ('ADMIN','MENTOR','LEARNER')	Role granted on accept
token_hash	text	NO	UNIQUE	SHA-256 of the invite token
invited_by	uuid	YES	FK users(id) ON DELETE SET NULL	Inviter
created_at	timestamptz	NO	default now()	Row creation time
expires_at	timestamptz	NO	-	Invite expiry
accepted_at	timestamptz	YES	-	Set when accepted

Indexes and constraints: INDEX (organization_id, email).

3.3 Skill graph

tracks [S1]

A learning track such as Backend Python. Only one track needs content in Sprint 1.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
slug	text	NO	UNIQUE	e.g. 'backend-python'
name	text	NO	-	Display name
description	text	YES	-	Short description for the track card
icon	text	YES	-	Icon name or emoji for the card
status	text	NO	default 'COMING_SOON'; CHECK IN ('ACTIVE','COMING_SOON','HIDDEN')	ACTIVE tracks can be tested; COMING_SOON are greyed out
display_order	smallint	NO	default 0	Sort order on the track screen
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)

skills [S1]

Nodes of the skill DAG.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
track_id	uuid	NO	FK tracks(id) ON DELETE CASCADE	Track the skill belongs to
slug	text	NO	UNIQUE with track_id	Stable key used in seed files

Column	Type	Null	Default / constraint	Description
name	text	NO	-	Display name, e.g. 'REST API design'
cluster	text	NO	-	Topic group, e.g. 'databases'. Used to balance test questions.
difficulty	numeric(4,2)	NO	-	On the same scale as ability (theta), about -3 to 3. Used to mark mastery.
description	text	YES	-	What the skill covers
display_order	smallint	NO	default 0	Sort order within cluster
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)

Indexes and constraints: UNIQUE (track_id, slug); INDEX (track_id).

skill_prerequisites [S1]

Edges of the DAG: prerequisite_id must be learned before skill_id.

Column	Type	Null	Default / constraint	Description
skill_id	uuid	NO	PK part; FK skills(id) ON DELETE CASCADE	The dependent skill
prerequisite_id	uuid	NO	PK part; FK skills(id) ON DELETE CASCADE	The skill that must come first

Indexes and constraints: PRIMARY KEY (skill_id, prerequisite_id); CHECK (skill_id <> prerequisite_id); INDEX (prerequisite_id). Cycles are rejected by the application with a topological sort.

3.4 Question bank

assessment_items [S1]

Multiple-choice questions with 3-parameter IRT values.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
track_id	uuid	NO	FK tracks(id) ON DELETE CASCADE	Track the question belongs to
stem	text	NO	-	Question text
options	jsonb	NO	-	Array of answer strings, e.g. ["A","B","C","D"]
correct_option	smallint	NO	-	Zero-based index into options
explanation	text	YES	-	Why the answer is correct (for later review screens)
cluster	text	NO	-	Main topic cluster; used for content balancing during selection
irt_a	numeric(5,3)	NO	default 1.000; CHECK > 0	Discrimination
irt_b	numeric(5,3)	NO	-	Difficulty, same scale as theta
irt_c	numeric(4,3)	NO	default 0.200; CHECK >= 0 AND < 1	Guessing probability
calibration_source	text	NO	default 'EXPERT'; CHECK IN ('EXPERT','EMPIRICAL')	Where a, b, c came from. EXPERT until recalibrated from real answers.
status	text	NO	default 'DRAFT'; CHECK IN ('DRAFT','ACTIVE','RETIRED')	Only ACTIVE questions are served
exposure_count	integer	NO	default 0	Times served; used for the exposure cap

Column	Type	Null	Default / constraint	Description
times_answered	integer	NO	default 0	Total answers (for recalibration)
times_correct	integer	NO	default 0	Total correct answers (for recalibration)
created_by	uuid	YES	FK users(id) ON DELETE SET NULL	Author
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)

Indexes and constraints: INDEX (track_id, status); INDEX (track_id, cluster).

item_skills [S1]

Which skills a question tests.

Column	Type	Null	Default / constraint	Description
item_id	uuid	NO	PK part; FK assessment_items(id) ON DELETE CASCADE	Question
skill_id	uuid	NO	PK part; FK skills(id) ON DELETE CASCADE	Skill tested

Indexes and constraints: PRIMARY KEY (item_id, skill_id); INDEX (skill_id).

3.5 Assessments

assessments [S1]

One row per diagnostic test attempt.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
user_id	uuid	NO	FK users(id) ON DELETE CASCADE	Learner
organization_id	uuid	YES	FK organizations(id) ON DELETE SET NULL	Organization context, if any
track_id	uuid	NO	FK tracks(id)	Track being tested
status	text	NO	default 'IN_PROGRESS'; CHECK IN ('IN_PROGRESS','COMPLETED','ABANDONED')	Attempt state
engine_version	text	NO	-	Version of the CAT engine used; needed to reproduce results
config	jsonb	NO	-	Settings frozen at start: prior, SE threshold, min and max items, grid
theta	numeric(6,4)	YES	-	Final ability estimate
theta_se	numeric(6,4)	YES	-	Standard error of the estimate (the 'confidence')
ci_low	numeric(6,4)	YES	-	Lower end of 95% interval
ci_high	numeric(6,4)	YES	-	Upper end of 95% interval
level	text	YES	CHECK IN ('BEGINNER','INTERMEDIATE','ADVANCED')	Label derived from theta
items_administered	smallint	NO	default 0	Questions answered so far
correct_count	smallint	NO	default 0	Correct answers so far

Column	Type	Null	Default / constraint	Description
stop_reason	text	YES	CHECK IN ('SE_THRESHOLD','MAX_ITEMS','NO_ITEMS_LEFT')	Why the test ended
started_at	timestamptz	NO	default now()	Start time
last_activity_at	timestamptz	NO	default now()	Latest answer time; used to expire abandoned tests
completed_at	timestamptz	YES	-	End time

Indexes and constraints: INDEX (user_id, started_at DESC); UNIQUE (user_id, track_id) WHERE status = 'IN_PROGRESS' (one active attempt per learner per track).

assessment_responses [S1]

Ordered log of every answer. Replaying it reproduces theta and SE exactly.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
assessment_id	uuid	NO	FK assessments(id) ON DELETE CASCADE	Attempt
seq	smallint	NO	-	1-based order of the question in this test
item_id	uuid	NO	FK assessment_items(id)	Question served
selected_option	smallint	NO	-	Option index the learner chose
is_correct	boolean	NO	-	Whether the answer was right
response_time_ms	integer	YES	-	Time taken to answer
theta_after	numeric(6,4)	NO	-	Ability estimate after this answer
se_after	numeric(6,4)	NO	-	Standard error after this answer
answered_at	timestamptz	NO	default now()	Answer time

Indexes and constraints: UNIQUE (assessment_id, seq); UNIQUE (assessment_id, item_id); INDEX (item_id) for later recalibration.

assessment_skill_results [S1]

Skill map produced by the mapper at the end of a test.

Column	Type	Null	Default / constraint	Description
assessment_id	uuid	NO	PK part; FK assessments(id) ON DELETE CASCADE	Attempt
skill_id	uuid	NO	PK part; FK skills(id) ON DELETE CASCADE	Skill
state	text	NO	CHECK IN ('MASTERED','FRONTIER','LOCKED')	MASTERED = known, FRONTIER = ready to learn, LOCKED = prerequisites missing

Indexes and constraints: PRIMARY KEY (assessment_id, skill_id).

3.6 Async AI tasks

ai_tasks [Later (table now)]

Gateway-side record of each job handed to the Node worker.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id

Column	Type	Null	Default / constraint	Description
user_id	uuid	NO	FK users(id) ON DELETE CASCADE	Who asked for the task
organization_id	uuid	YES	FK organizations(id) ON DELETE SET NULL	Organization context, if any
type	text	NO	-	ROADMAP_GENERATE, RESULT_EXPLAIN, ITEM_DRAFT, SKILL_GRAPH_DRAFT
status	text	NO	default 'PENDING'; CHECK IN ('PENDING','RUNNING','DONE','FAILED','CANCELLED')	Task state
priority	smallint	NO	default 5	Lower number runs first
input	jsonb	NO	-	Input sent to the worker
result	jsonb	YES	-	Validated output JSON
schema_version	text	YES	-	Version of the /contracts schema the result follows
progress	smallint	NO	default 0; CHECK BETWEEN 0 AND 100	Percent shown to the user
attempts	smallint	NO	default 0	Tries so far
max_attempts	smallint	NO	default 3	Retry limit
error_code	text	YES	-	LLM_TIMEOUT, INVALID_OUTPUT, etc.
error_message	text	YES	-	Human-readable failure detail
idempotency_key	text	YES	UNIQUE with user_id	Stops duplicate submissions
worker_job_id	text	YES	-	Queue job id returned by the worker
created_at	timestamptz	NO	default now()	Row creation time
started_at	timestamptz	YES	-	When the worker began
finished_at	timestamptz	YES	-	When it finished or failed

Indexes and constraints: INDEX (user_id, created_at DESC); INDEX (status) WHERE status IN ('PENDING','RUNNING'); UNIQUE (user_id, idempotency_key).

3.7 Roadmaps and progress (planned for Sprint 2)

roadmaps [Later]

A generated learning roadmap for one learner and track.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
user_id	uuid	NO	FK users(id) ON DELETE CASCADE	Owner
track_id	uuid	NO	FK tracks(id)	Track
assessment_id	uuid	YES	FK assessments(id) ON DELETE SET NULL	Test the roadmap was based on
task_id	uuid	YES	FK ai_tasks(id) ON DELETE SET NULL	Task that generated it
title	text	NO	-	Roadmap title
summary	text	YES	-	Short overview
status	text	NO	default 'ACTIVE'; CHECK IN ('ACTIVE','ARCHIVED')	State
schema_version	text	NO	-	Contract version of the stored JSON

Column	Type	Null	Default / constraint	Description
content	jsonb	NO	-	Full validated LLM output, kept for re-rendering
created_at	timestamptz	NO	default now()	Row creation time
updated_at	timestamptz	NO	default now()	Last change time (set by the application on update)

Indexes and constraints: INDEX (user_id, status).

roadmap_steps [Later]

Individual steps of a roadmap, tracked by the learner.

Column	Type	Null	Default / constraint	Description
id	uuid	NO	PK, default gen_random_uuid()	Unique id
roadmap_id	uuid	NO	FK roadmaps(id) ON DELETE CASCADE	Parent roadmap
skill_id	uuid	YES	FK skills(id) ON DELETE SET NULL	Skill this step teaches
seq	smallint	NO	-	Order within the roadmap
title	text	NO	-	Step title
description	text	YES	-	What to learn or do
resources	jsonb	NO	default '[]'	Links and materials: [{ title, url, type }]
estimated_hours	numeric(5,1)	YES	-	Estimated effort
status	text	NO	default 'NOT_STARTED'; CHECK IN ('NOT_STARTED','IN_PROGRESS','DONE')	Learner progress
completed_at	timestamptz	YES	-	When marked done

Indexes and constraints: UNIQUE (roadmap_id, seq).

user_skill_progress [Later]

Current standing of a learner on each skill (updated by tests and by completed steps).

Column	Type	Null	Default / constraint	Description
user_id	uuid	NO	PK part; FK users(id) ON DELETE CASCADE	Learner
skill_id	uuid	NO	PK part; FK skills(id) ON DELETE CASCADE	Skill
state	text	NO	CHECK IN ('MASTERED','FRONTIER','LOCKED')	Current state
source	text	NO	CHECK IN ('ASSESSMENT','MANUAL')	What last changed it
source_assessment_id	uuid	YES	FK assessments(id) ON DELETE SET NULL	Test that set it, if any
updated_at	timestamptz	NO	default now()	Last change

Indexes and constraints: PRIMARY KEY (user_id, skill_id).

Part 4. Table relationships

Child table . column	Parent table . column	On delete	Meaning
refresh_tokens.user_id	users.id	CASCADE	A user's sessions
password_reset_tokens.user_id	users.id	CASCADE	A user's reset tokens
organization_members.organization_id / user_id	organizations.id / users.id	CASCADE	Membership
organization_invites.organization_id	organizations.id	CASCADE	Pending invites
skills.track_id	tracks.id	CASCADE	Skills of a track
skill_prerequisites.skill_id / prerequisite_id	skills.id	CASCADE	DAG edges
assessment_items.track_id	tracks.id	CASCADE	Questions of a track
item_skills.item_id / skill_id	assessment_items.id / skills.id	CASCADE	Question to skill tags
assessments.user_id	users.id	CASCADE	A learner's attempts
assessments.track_id	tracks.id	none	Track tested
assessment_responses.assessment_id	assessments.id	CASCADE	Answers of an attempt
assessment_responses.item_id	assessment_items.id	none	Question served
assessment_skill_results.assessment_id / skill_id	assessments.id / skills.id	CASCADE	Skill map of an attempt
ai_tasks.user_id	users.id	CASCADE	A user's tasks
roadmaps.user_id / track_id	users.id / tracks.id	CASCADE / none	Roadmap owner and track
roadmaps.assessment_id / task_id	assessments.id / ai_tasks.id	SET NULL	Origin of a roadmap
roadmap_steps.roadmap_id	roadmaps.id	CASCADE	Steps of a roadmap
roadmap_steps.skill_id	skills.id	SET NULL	Skill a step teaches
user_skill_progress.user_id / skill_id	users.id / skills.id	CASCADE	Learner standing per skill